POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Fall

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is buffer solution? Write the mechanism of buffer action of acidic buffer with suitable example. Calculate the pH of buffer solution containing 400 ml of 0.3 M acetic acid and 200 ml of 0.6 M sodium acetate (K_a of acetic acid is 1.8×10^{-5}) 8

   b) Differentiate between chemical corrosion and electrochemical corrosion. From the given electrode potential values answer the following : 7

   i. Write the electrode reaction.

   ii. Write the cell reaction.

   iii. Calculate the e.m,f. of the cell at 298K when both the electrodes are coupled together .

   $E^0 Zn^{2+}/Zn = -0.76V$

   $E^0 Ag^+/Ag = +0.80\ V$

   $R = 8.314\ J\ mol^{-1}\ K^{-1}$

   $F = 96500C$

   $[Zn^{2+}] = 0.01M$ $[Ag^+] = 0.1M$

2. a) Explain the following for the transition elements: 8

   i. form colored compounds

   ii. paramagnetic in nature

   b) What is electron affinity? Write the factors that affect electron affinity. Between nitrogen and oxygen which one has higher ionization energy give reason? 7

3. a) Explain the mechanism and stereochemistry of SN_2 reaction. 8

   b) What is electrophilic aromatic substitution? Write the mechanism of Halogenation taking suitable example. 7

4. a) Differentiate between thermoplastic and thermosetting plastics. Write the preparation, properties and uses of 8

   i. Teflon
   ii. Polyester
   iii. PVC

   b) What is a chemical composition of natural rubber? How is it processed to obtain crepe rubber and smoked rubber? Write some advantages of vulcanized rubber. 7

5. a) What explosive describe the different types of explosive. Also write the preparation and uses of TNT and TNG 8

   b) Explain briefly: 7

   i. Paper chromatography
   ii. Mass Spectroscopy

6. a) What is hard water? Write its types and also methods of removal. 8

   b) Define alkalinity. How it can be measured in the laboratory? Describe it. 7

7. Write short notes on: **(Any two)** 2×5

   a) Biological oxygen demand
   b) SN_1
   c) NMR Spectroscopy